

VARIABLES, IN PARTICULAR RANDOM VARIABLES

Abstract. We argue that the concept of a variable as classically understood in mathematics and physics is didactically and scientifically inadequate for the needs of probability and statistics and we propose an alternative concept. A revision should have political implications for the relation between mathematical and statistical teaching.

Key words: axiomatics, foundations of probability, random variables, statistical teaching, variables.

It can hardly be doubted that the concept of a variable is fundamental and ubiquitous in analysis. The textbooks are very reserved with explanations or definitions, however. On the other hand, they take great pain in explaining and defining the concept of a function, a concept which is clearly secondary to the concept of a variable. The concept of a function (or a mapping) builds on the concept of variables; a function associates the value of the dependent variable with the value of the independent variable.

I find it remarkable that the textbooks do not say or suggest, that the concept of a variable may remain undefined just as the concept of a point in euclidean geometry. They rather seem to suggest that it must be left to the users of analysis to build up that concept of a variable which suits their needs. I am not aware of any didactical or philosophical discussion of this kind of an approach to mathematics, however.

At first sight the didactical strategy seems to work in present day elementary analysis. Different contexts go along with different conceptions of variables. Nobody is really bothered by the observation that variables in thermodynamics are not exactly the same as variables in mechanics or economics. I will argue here, that the disconcert with a mathematical concept of a variable has shown its Achilles-heal in the case of random variables. It seems to me, that present day education in elementary analysis is not open for an intuitive and workable conception of a random variable. This has lead to a very unfortunate situation which cries for action.

A very basic innovation is needed to pave the way for a broader and deeper understanding of stochastics. Mathematicians must understand that the common reductionist approach to the concept of a random variable is inadequate and that this must have consequences for the teaching of elementary analysis already at school level.

There are of course historical reasons for the divergence between the needs of stochastics in the practical sense and present day mathematics in the narrow sense. When probability around 1930 (after a dormant period of a century) became again a respectable field of mathematics, the teachers of mathematics wanted a definition: what is a random variable? Bourbaki did not offer a definition, he left stochastics aside. But for the mathematicians who believed in Bourbaki at that time it was clear,

that a random variable is a measurable function (or a measurable mapping) from a measurable space $(\Omega, \mathcal{A})$ into another measurable space $(S, \mathcal{B})$. And they did not try to please traditional stochastic thinking. P. Halmos argued in these times that probability theory is legitimate mathematics, in as far that it is the theory of normed boolean algebras.

The mathematics departments have put probability theory into their curricula, because they were told in the sixties that mathematicians need competency in stochastics, when they were to leave the realm of genuine mathematics, in other words, when they leave the mathematics department. By this move the advanced students of mathematics towards the end of their studies after an intensive training in traditional analysis got the opportunity to learn the reductionist approach to stochastics. The majority of university mathematicians considered this as a concession which was not justified by the strive for mathematical progress.

On the other hand, the concession turned out to be a fairly small contribution to the needs of a better stochastical education. Only very few students got acquainted with the rich world of stochastic intuitions. Not much was achieved to assist students of the sciences working with unprecise data or students of economics dealing with decisions under uncertainty. A mathematics based philosophy of stochastical thinking didn't come to the attention of any kind of students. A gap was opened: the non-mathematicians defied measure theory, and the mathematicians defied "cookbook"-statistics, as they called stochastics without measure-theoretic foundations. In particular, mathematicians would not accept the "definition" of a random variable, which is commonly presented in books for non-mathematicians: "*A random variable is a quantity whose value depends on chance*".

More and more people have stressed in recent years that action has to be taken in order to provide a common ground for mathematical stochastics and stochastical modelling in the sciences. There is a remarkable article of two prominent statisticians: David S. Moore and George W. Cobb (2000). The introduction says:

It has become a truism, at least among statisticians, that while statistics is a mathematical science, it is not a subfield of mathematics ... In mathematics, context obscures structure. In data analysis, context provides meaning. ...

They conclude their article:

Mathematics, a core discipline, looks inward and risks being seen as increasingly irrelevant. Statistics, a methodological discipline, looks outward but risks being swallowed by information technology. Both professions have a stake in the survival of statistics as a subject informed and structured by mathematics.

The article also points out political consequences of a reconciliation:

To mathematics statistics offers not only the example of an outward looking culture, but also entree to new problems ripe for mathematical study. To statistics, mathematics offers not only the safe harbor of organizational strength, but intellectual anchorage as well: mathematical understanding is an essential part of what distinguishes statistical thinking from most of the rest of information technology.

Mathematics is challenged both didactically-philosophically and politically in these days; and stochastics will undoubtedly be a focus of renewal. There are pleas in this direction also by prominent mathematicians.

David Mumford (former IMU-president, originally coming from the very pure end of mathematics) contributed to the volume “Mathematical Frontiers and Perspectives 2000” an article “The Dawning of the Age of Stochasticity.” There he points out:

The basic object of study in probability is the random variable, and I will argue that it should be treated as a basic construct, like spaces, groups and functions, and it is artificial and unnatural to define it in terms of measure theory.

Mumford challenges the teachers of probability:

Put the concept of ‘random variable’ on center stage and work with manipulations of random variables wherever possible.

The challenge has been felt by many teachers of stochastics for years. In our courses at the Department of Mathematics at Frankfurt University, Goetz Kersting and myself have been working hard over 20 years to deal with random variables independently of measure theory. In more recent times we have got encouraging results both in technical aspects and in practical teaching. We feel that the successes of teaching have considerably improved since the curriculum advises the students of mathematics to take up stochastics early. In former times we had to deal with the problem that students with an extensive training in traditional analysis were disappointed or confused, when the course in stochastics didn’t conform to the familiar scheme. These students insisted that they would understand much more easily on the basis of measure theory; there was little curiosity for stochastic intuition. Our younger students now find it less difficult to build up intuitions around the concept of random variables. In our elementary courses this concept is introduced in a way, which is open for enlargement fitting the needs of professional mathematical stochastics.

In the search for a flexible conception of a variable it is appropriate to recall the historical origins of analysis. It was Leibniz who invented functions. Newton on the other hand had curves in mind when he invented his calculus. Euler (1768) considered it fruitless to unite the conceptions in Newton’s tradition with the ideas of the continental school. In early 19th century a variable was commonly understood as something which runs through a domain. It was in opposition to this dynamical conception when Bolzano gave a statical definition of continuity. This led to the concept of a function “in the sense of Dirichlet” and finally to Cantor’s set theory, forbidding any dynamical association.

The dynamical conception of a variable has survived in school mathematics. Unfortunately there it is so intimately connected with special functions on an interval that every extension of the intuition about variables seems to be difficult. (Already Riemann’s idea of an abstract “multiply extended manifold” seems to be beyond the intuition of most of our high-school teachers). Thus there is very little hope that an intuitive conception of random variables might profit from the traditional concept of a variable in analysis. As far as I know, nobody has ever tried to teach an intuitive concept of a random variable using analogies with the algebraic concept of an indeterminate. Thus random variables have to find their own intuitive basis.

Modern analysis in the spirit of Cantor’s set theory has variables of course; but nothing is varying here. The mathematician may insert a value of the independent

“variable” into a function $f(\cdot)$ (which is seen as a black box) and he retains the value of the dependent “variable”; the evaluation is a singular act at every instance; the mathematician is choosing a value in an arbitrary, completely discontinuous act. For every singular x in a specified set the black box $f(\cdot)$ produces a singular $y = f(x)$.

In the case of a random variable we have an actual variability and this variability is due to chance. Chance inserts a “typical” value into the black box $f(\cdot)$ and the output is a “typical” value of the dependent variable.

$$X \xrightarrow{f} Y = f(X).$$

The technical term is not “typical”, but rather “random”. “Typical value” might be confused with “generic value” which is a technical term in geometry. So we say: The blackbox associates the random variable $Y = f(X)$ with the random variable X .

In stochastics the calculus of functions is more or less the same as in traditional analysis. The arguments of the functions have a different logical status, however. There are instances where the difference has technical consequences. For example: assume $Y = f(X)$. We may ask for conditions which guarantee the existence of a $g(\cdot)$ such that $X = g(Y)$. (In our “polish” calculus to be sketched below there exists an easy and intuitive criterion).

There exist teaching traditions for elementary statistics (or probability) in various applied fields. They all deal with random variables. And most of them seem to get along without any elaborate mathematics. One may object that all of them are fairly incoherent. But this is not the most serious objection. Looking closer one finds that the non-mathematical traditions tend to produce misconceptions which hamper a deeper understanding of stochastic modelling. For example: in certain traditions of teaching randomized algorithms, the concept of randomness (implicitly) includes stochastic independence or uniform distribution or both.

Almost all traditions of teaching elementary probability theory tie the concept of a random variable to a fixed “true” distribution. Many textbooks stick to the idea that the distribution is the only thing which is worthwhile considering when we speak about random variables; there, probability theory is the theory of probability measures. This has its price. We find that very often students who have been brought up in a narrow teaching tradition, have serious problems, when they have to think in wider terms. For example, the concept of an equivalent martingal measure, which is crucial in mathematical finance, presents enormous conceptual difficulties to most of these students, as they are used to think in terms of the familiar “true” distribution.

Let me now briefly sketch our proposal, how random variables might be conceived as objects within an axiomatic context. We start with four axioms which can be supplemented at various levels all compatible with professional mathematical stochastics.

- Every random variable takes values in a measurable space (“domain”). When X_1 and X_2 are random variables, then (X_1, X_2) is a random variable with values in the product space.
- When X is an S_1 -valued random variable and $\varphi: S_1 \rightarrow S_2$ is a measurable mapping, then $\varphi(X)$ is an S_2 -valued random variable.
- For every measurable B in the domain of X there exists a $\{0, 1\}$ -valued random variable $1_{\{X \in B\}}$ which represents the event $\{X \in B\}$. The S -valued random variables X and Y are equal if and only if the event $\{X \in B\}$ is equal to the event $\{Y \in B\}$ for every measurable B (or a family of subsets which generates the σ -algebra).
- The set $\mathcal{A}$ of observable events is a σ -complete boolean algebra (Equality of events is given axiomatically and cannot be reduced to equality of more concrete mathematical objects).

It is true that the events can be identified with the $\{0, 1\}$ -valued random variables. However the universe of all random variables (in any particular stochastic model) is much richer than the set of those random variables which can be constructed in an obvious way from the $\{0, 1\}$ -valued random variables. Even in school stochastics, we need nondiscrete random variables, in particular random variables with values in $\mathbb{R}$, and random vectors, i. e. random variables with values in $\mathbb{R}^n$. In elementary university teaching we need random functions and random measures. Ultimately our axiomatic system admits random variables with values in all measurable spaces $(S, \mathcal{B})$ of a certain type $\mathcal{S}$. This type $\mathcal{S}$ has to be specified and I may say at this point, that my favorite class $\mathcal{S}$ is the class of polish spaces.

Beyond the four axioms stated above, one has to make a couple of non-obvious technical and didactical decisions. The teacher has to decide how far he wants to extend the perspective at a certain point – but he should be aware of what he is doing and he should be careful not to brain-wash his students at a level which cannot be extended in an intuitive way.

We believe that an essential point of stochastic reasoning would be missed, if one would teach a toy version which would require all the admissible domains to be denumerable. Such a toy version would be counterintuitive for most (high-school) students, since the students at the same time might be confronted with random variables in the sense of unprecise measurements in some applied discipline.

Philosophically speaking, such a toy version would miss a “principle of continuity” (in a sense which I have learned in long discussions with Michael Otte). We insist, that it is basic for a thinking in stochastic terms, that the events (in general) cannot be broken down to any kind of “elementary events” (which in the measure-theoretic reduction would correspond to the values of some independent variable $\omega \in \Omega$). There are school teachers who have tried to tell their students what elementary events are in the case of an infinite sequence of coin tosses. They all have failed; and we argue that there are fundamental philosophical reasons for this failure. Events are not like sets, which are equal if and only if they have the same elements;

the equality of events is more abstract and flexible. The partial analogy between sets and events is the philosophical trap which every teacher of stochastics should be aware of. (Of course the advanced analyst will be happy to learn that there exists a technical although highly non-constructive bridge to set theory. 1948 Loomis has proved that every σ -complete Boolean lattice can be represented by a set of equivalence classes of measurable sets.)

Here is an example of a nontrivial didactical decision. At some instance the system of axioms has to say under what conditions an infinite sequence of random variables $X_1, X_2, \dots$ may itself be considered as a random variable in the universe. In particular it has to be fixed axiomatically under what conditions an infinite product of admissible spaces $(S_i, \mathcal{B}_i)$ can be considered as an admissible space. We think that an infinite sequence of coin tosses or a random path through a rooted tree ought to qualify as a random variable at a fairly early stage of stochastic education.

For a second course at university level I strongly recommend the full class of polish spaces. This opens a wide range of interesting questions and fruitful considerations which fortunately have nice intuitive answers. It turns out, that the theory of “sure” convergence of random variables is more or less a well reflected version of the material of convergence and continuity, which is commonly treated in elementary analysis. The main ingredients of sophistication are the following: we experience the distinction between sure events and mathematically true assertions. Secondly, all manipulations of the objects are bound to be of denumerable type. Thirdly the considerations about nullsets in measure theory get a clear profile. Finally the floor is set for the theory of weak convergence (which is a subject for advanced courses). It gains a nice intuitive background by the concept of sure convergence; stochastic convergence appears as the topological weakening of sure convergence.

This is not the place to say more about mathematical technique. A thorough scrutiny of the technical aspects of our axiomatization is in preparation. On Kersting’s homepage you can find a manuscript documenting a first university course which puts random variables on center stage. More about tensions between stochastics and mathematics proper can be found in Dinges (2001).

As a conclusion I may summarize the main thrust of this article in honour of my dear friend Michael: Students on all levels ought to be encouraged to trust in the intuitive concept of a random variable. The variability of an observable may be due to chance; it is not necessarily caused by the specification of some “independent” variable.

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